Phys 115B HW 8

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1 D

Question 1. *Consider a system with 2 N-state systems, with $N \gg 1$. For example, for the case where $N = 2$, you could imagine two spin 1/2 particles, or for a case with $N = 100$, two particles in a 1-D harmonic oscillator but confined to the lowest 100 levels. The point is you have only two particles, but they can fill up to N levels. The system is assumed to be warm compared to ΔE, so all states are occupied equally.*

(a) What is the probability for finding the particles in the same state if they are distinguishable?

(b) What is the probability for finding the particles in the same state if they are fermions?

(c) What is the probability for finding the particles in the same state if they are bosons?

Proof. Let's fix the notation, say $\mathcal{H}$ is the N-dimensional Hilbert Space describing the N-state system, and $|\psi_1\rangle, ..., |\psi_N\rangle$ are the allowed states upon observables.

(a) If the two particles are distinguishable, then the Hilbert Space describing it is $\mathcal{H} \otimes \mathcal{H}$, so total of N^2-dimensional. Which, there are N^2 distinct states (i.e. all $|\psi_i\rangle \otimes |\psi_j\rangle$, where $i, j \in \{1, ..., N\}$), while N of them have both particles being in the same states (i.e. all $|\psi_i\rangle \otimes |\psi_i\rangle$, where $i \in \{1, ..., N\}$). Hence, with each state being equally probable to observe, the probability the two distinguishable particles are in a same state is $\frac{N}{N^2} = \frac{1}{N}$.

(b) If the two particles are fermions, they're not allowed to be in the same state, hence the robability of finding two fermions in the same state is 0.

(c) If the two particles are bosons, then the Hilbert Space describing it is $S^2\mathcal{H}$ the symmetric tensor subspace in $\mathcal{H} \otimes \mathcal{H}$. Which, there are $\frac{N(N+1)}{2}$ distinct states (i.e. all $|\psi_i\rangle \otimes |\psi_j\rangle + |\psi_j\rangle \otimes |\psi_i\rangle$ for $i, j \in \{1, ..., N\}$), while N of them have both particles being in the same states (i.e. all $|\psi_i\rangle \otimes |\psi_i\rangle$). Hence, with each state being equally probable to observe, the probability the two bosons are in a same state is $\frac{N}{N(N+1)/2} = \frac{2}{N+1}$.

□

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Question 2. *Consider a free electron gas with unequal numbers of spin-up and spin-down particles (N_+ and N_- respectively). Such a gas would have a net magnetization (magnetic dipole moment per unit volume)*

$$M = -\frac{N_+ - N_-}{V}\mu_B \hat{k} = M\hat{k}$$

where $\mu_B = \frac{e\hbar}{2m_e}$ is the Bohr magneton (The minus sign is there because the charge of the electron is negative).

(a) Assuming the electrons occupy the lowest energy levels consistent with the number of particles in each spin orientation, find E_{tot}. Check that your answer reduces to

$$E_{tot} = \frac{\hbar^2(3\pi^2)^{5/3}(2N)^{5/3}}{10\pi^2 m}V^{-2/3}$$

when $N_+ = N_- = N$.

(b) Show that for $M/\mu_B \ll \rho \equiv (N_+ + N_-)/V$ (which is to say $|N_+ - N_-| \ll (N_+ + N_-)$), the energy density is

$$\frac{1}{V}E_{tot} = \frac{\hbar^2(3\pi^2\rho)^{5/3}}{10\pi^2 m}\left[1 + \frac{5}{9}\left(\frac{M}{\rho\mu_B}\right)^2\right].$$

The energy is a minimum for $M = 0$, so the ground state will have zero magnetization However, if the gas is placed in a magnetic field (or in the presence of interactions between the particles) it may be energetically favorable for the gas to magnetize.

Proof.

(a) Let $N_{\text{tot}} := N_+ + N_-$, and $N_{\text{com}} := \min\{N_+, N_-\}$ (so, how many electrons in total, and how many pairs of opposit spin electrons, respectively). Another way to represent the numbers is $N_{\text{com}} = N_{\text{tot}}$

Since each energy level allows two electrons with opposite spins, it's the most favoarable for the lowest energy levels to be filled with

(b)

$\square$

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Question 3. *The free electron gas model ignores the Coulomb repulsion between electrons. Because of the exchange force, Coulomb repulsion has a stronger effect on two electrons with anti- parallel spins (which behave in a way like distinguishable particles) than two electrons with parallel spins (whose position wave function must be antisymmetric - we saw this with the carbon atom). As a crude way to take into account Coulomb repulsion, pretend that every pair of electrons with opposite spin carry extra energy* U*, while electrons with the same spin do not interact at all; this adds* $\Delta E = UN_+N_-$ *to the total energy of the electron gas. As you will show, above a critical value of* U*, it becomes energetically favorable for the gas to spontaneously magnetize* ($N_+ \neq N_-$)*; the material becomes ferromagnetic.*

(a) Rewrite ΔE *in terms of the density* ρ *and the magnetization* M *from the previous problem.*

(b) Assuming that $M/\mu_B \ll \rho$*, for what minimum value of* U *is a non-zero magnetization energetically favored?*

Proof.

(a)

(b)

$\square$

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Question 4. *Show that for a Hermitian operator $\hat{Q}$, the operator $\hat{U} = e^{i\hat{Q}}$ is unitary. Hint: first prove that the adjoint is given by $\hat{U}^\dagger = e^{-i\hat{Q}}$; then prove that $\hat{U}^\dagger\hat{U} = 1$.*

Proof. Assume the exponentiation converges absolutely point wise (and that adjoint commutes with summation), then with $\hat{Q}$ being self adjoint (Hermitian), one have the following:

$$\hat{U} = e^{i\hat{Q}} = \sum_{n=0}^{\infty} \frac{1}{n!}(i\hat{Q})^n \tag{1}$$

In particular, the adjoint is given as follow:

$$\hat{U}^\dagger = \left(\sum_{n=0}^{\infty} \frac{1}{n!}(i\hat{Q})^n\right)^\dagger = \sum_{n=0}^{\infty} \frac{1}{n!}((i\hat{Q})^\dagger)^n = \sum_{n=0}^{\infty} \frac{1}{n!}(-i\hat{Q})^n = e^{-i\hat{Q}} \tag{2}$$

Which, the two operators composes to the following:

$$\hat{U}^\dagger\hat{U} = \sum_{n=0}^{\infty}\sum_{m=0}^{\infty} \frac{1}{(n!)(m!)}(-i\hat{Q})^n(i\hat{Q})^m = \sum_{n=0}^{\infty}\sum_{k=0}^{n} \frac{1}{(k!)((n-k)!)}(-i\hat{Q})^k(i\hat{Q})^{n-k} \tag{3}$$

$$= \sum_{n=0}^{\infty}\left(\sum_{k=0}^{n} \frac{(-1)^k}{(k!)((n-k)!)}\right)(i\hat{Q})^n \tag{4}$$

Which, for $n = 0$, the coefficient $\sum_{k=0}^{n} \frac{(-1)^k}{(k!)((n-k)!)} = 1$; else if $n > 0$, notice the following factor:

$$\forall z \in \mathbb{C}, \quad ((-1) + z)^n = \sum_{k=0}^{n} \binom{n}{k}(-1)^k z^{n-k} = \sum_{k=0}^{n} \frac{n!(-1)^k}{k!\ (n-k)!} z^{n-k} \tag{5}$$

Which, take $z = 1$, one arrives at $0 = ((-1)+1)^n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!\ (n-k)!}$, hence with $n! \neq 0$, then $\sum_{k=0}^{n} \frac{(-1)^k}{k!\ (n-k)!} = 0$.

As a result, we have:

$$\hat{U}^\dagger\hat{U} = \sum_{n=0}^{\infty}\left(\sum_{k=0}^{n} \frac{(-1)^k}{(k!)((n-k)!)}\right)(i\hat{Q})^n = (i\hat{Q})^0 = 1 \tag{6}$$

Grinding out similar summation, one has $\hat{U}\hat{U}^\dagger = 1$, showing $\hat{U}^\dagger = \hat{U}^{-1}$, hence $\hat{U}$ is unitary. $\square$

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Question 5. *When we first introduced Dirac notation, we fixed the inner product of position and momentum eigenstates using our foreknowledge of the position-space representation of states of definite momentum, i.e. we asserted that*

$$\langle x|p\rangle = Ae^{ipx/\hbar}$$

where A is a normalization factor. Now we can derive this using the properties of the translation operator $T(a)$. If the state $|p\rangle$ is one of well-defined momentum p, how would you characterize state $|p'\rangle = T(a)|p\rangle$? Show that the position-space wavefunctions of these states are related by

$$\langle x|p'\rangle = \psi_{p'}(x) = e^{-iap/\hbar}\psi_p(x) = e^{-iap/\hbar}\langle x|p\rangle$$

and

$$\psi_{p'}(x) = \psi_p(x-a).$$

Then conclude that

$$\langle x|p\rangle = Ae^{ipx/\hbar}.$$

Proof. Assuming we know the relation $\langle x|\psi\rangle = \psi(x)$, and $\langle x| T(a) |\psi\rangle = \psi(x-a)$, while $T(a)^\dagger = T(a)^{-1} = T(-a)$. Also, one has $\hat{P}$ the momentum operator satisfies $T(a) = e^{-ia\hat{P}/\hbar} = \sum_{n=0}^{\infty} \frac{(-ia/\hbar)^n}{n!}\hat{P}^n$ (where $\hat{P}|p\rangle = p|p\rangle$ by definition of "definite momentum", which is an eigenvector of the momentum operator). Hence, one has the following:

$$\begin{aligned}
\langle x|p'\rangle = \langle x| \left(T(a)|p\rangle\right) &= \langle x| \left(\sum_{n=0}^{\infty} \frac{(-ia/\hbar)^n}{n!}\hat{P}^n |p\rangle\right) && (7)\\
&= \langle x| \left(\sum_{n=0}^{\infty} \frac{(-ia/\hbar)^n}{n!}p^n |p\rangle\right) = \langle x| \left(\sum_{n=0}^{\infty} \frac{1}{n!}\left(-\frac{iap}{\hbar}\right)^n |p\rangle\right) && (8)\\
&= \langle x| \left(e^{-iap/\hbar}|p\rangle\right) = e^{-iap/\hbar}\langle x|p\rangle && (9)
\end{aligned}$$

Hence, one derives that $\psi_p(x-a) = \psi_{p'}(x) = e^{-iap/\hbar}\psi_p(x)$ (given the definition $\langle x| T(a) |p\rangle = \psi_p(x-a)$). As a result, take $u := -a$ (where $a \in \mathbb{R}$ is arbitrary), and plugin $x = 0$, one has the following:

$$\psi_p(u) = \psi_p(0-a) = e^{-iap/\hbar}\psi_p(0) = \psi_p(0)e^{ipu/\hbar} \tag{10}$$

Hence, with $A = \psi_p(0)$, so $\psi_p(u) = Ae^{ipu/\hbar}$ for all $u \in \mathbb{R}$. $\square$